

Architectural Style Analysis

Pros: Clear separation of concerns, easy to understand

Cons: Performance overhead, tight coupling between layers, difficult to scale individual components

Suitability for Dropshipping Inventory & Order Orchestrator: Good, but scaling individual services can be difficult

Microservices

Pros: Independent scaling, fault isolation, independent deployment

Cons: Operational complexity, network latency, data consistency challenges

Suitability for Dropshipping Inventory & Order Orchestrator: Highly suitable

Client-Server

Pros: Centralized control, simple deployment, easy data consistency

Cons: Single point of failure, scalability bottleneck, limited fault tolerance

Suitability for Dropshipping Inventory & Order Orchestrator: Less suitable

Architecture Selection & Component Identification

Selected Architecture: Microservices Architecture

Components: 1. Order Manager Component

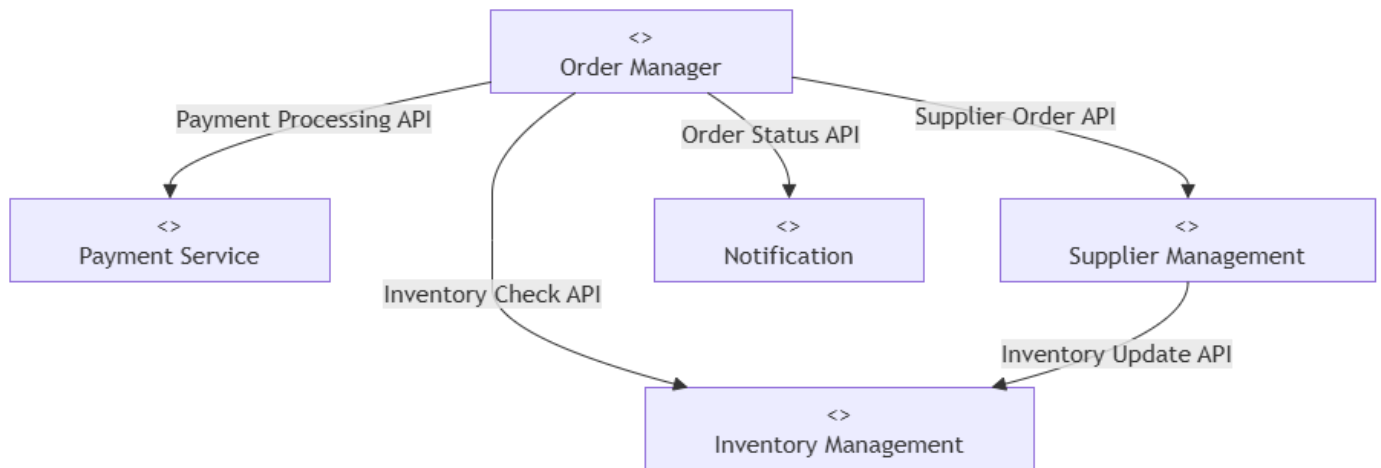
2. Payment Service Component

3. Inventory Management Component

4. Supplier Management Component

5. Notification Component

UML Diagram



Architecture Selection

We chose Microservices Architecture for the Dropshipping Inventory & Order Orchestrator System.

Architectural Choice:

Microservices Architecture was selected because the system contains multiple independent functions such as order management, payment processing, inventory management, supplier management, and notifications.

Reason 1:

Each service can be developed, deployed, and scaled independently according to its workload.

Reason 2:

Microservices provide fault isolation, so a failure in one supplier or service does not necessarily affect the entire system.

Security Advantage:

Each service can have its own access controls and authentication, reducing unnecessary access between system components.

Performance Benefit:

Individual services can be scaled independently, allowing high-load services such as order and inventory processing to handle more requests efficiently.

